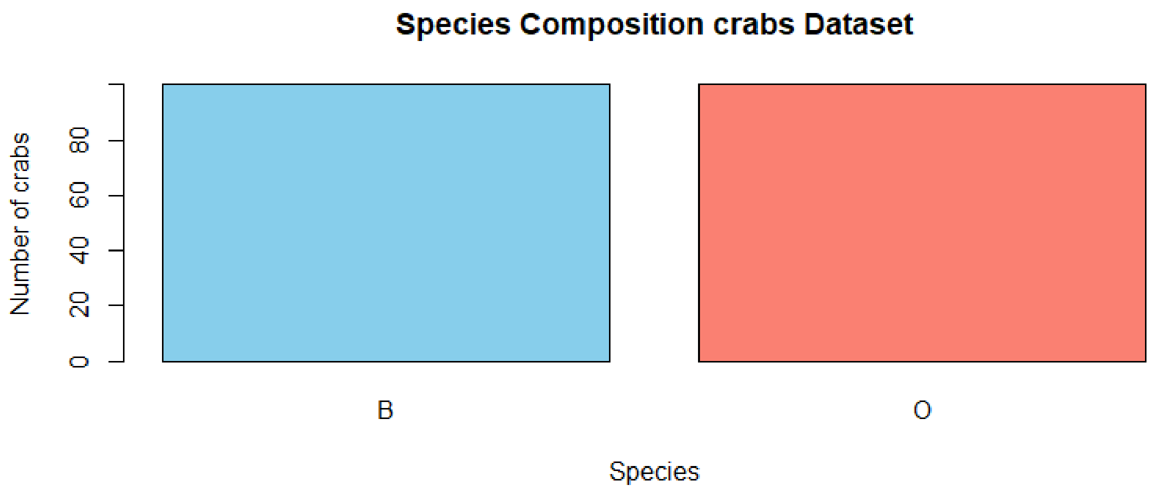
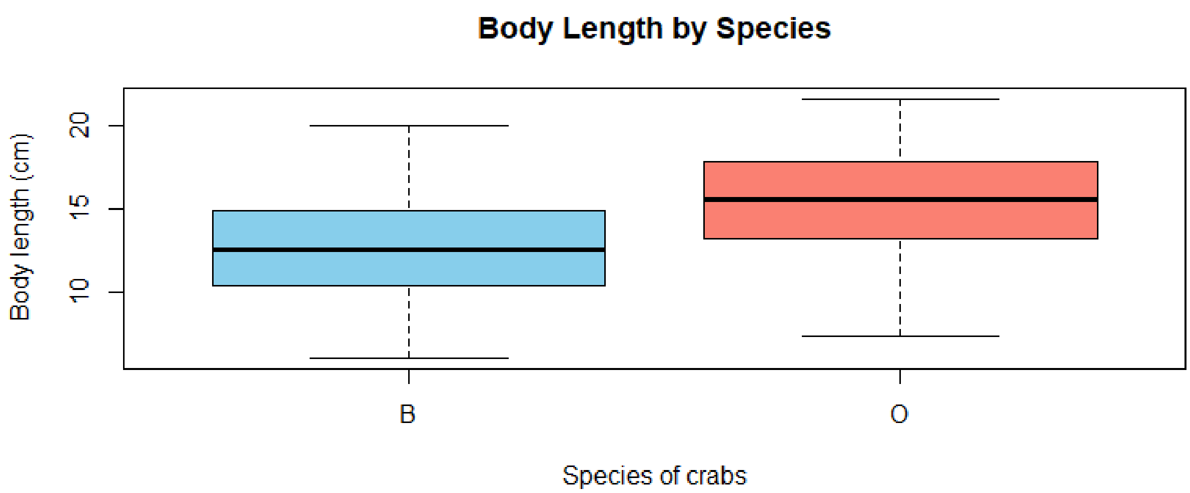
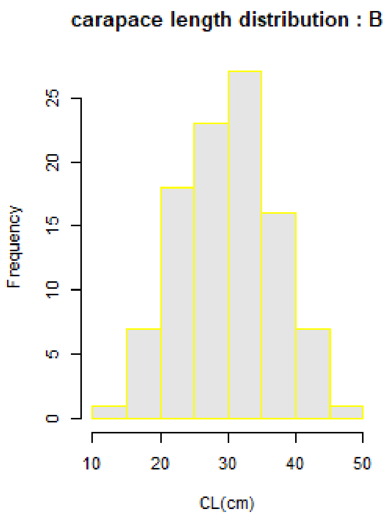
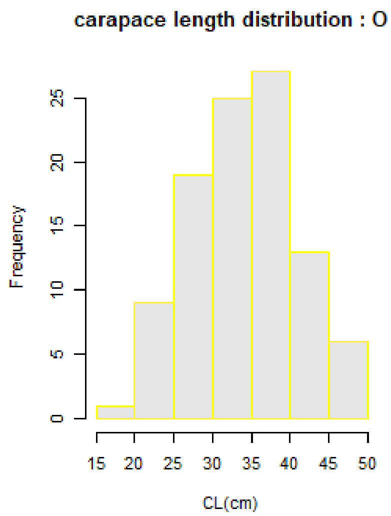
ASTA 607_22425133










```
data.frame':    200 obs. of  8 variables:
 $ sp   : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex  : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
 $ index: int  1 2 3 4 5 6 7 8 9 10 ...
 $ FL   : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
 $ RW   : num  6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL   : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
 $ CW   : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
 $ BD   : num  7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
> dim(crabs)
[1] 200   8
> sapply(crabs,class)
    sp     sex    index     FL     RW     CL     CW     BD
```

"factor" "factor" "integer" "numeric" "numeric" "numeric" "numeric" "numeric"

INTEGRATED STATISTICAL REPORT

```
####################################
# a) data set in Mass
library(MASS )
# b) data structure
data(crabs)
str(crabs)
dim(crabs)
sapply(crabs,class)

# c) frequency tables for species and sex
freq_tbl <- table(crabs$sp)
freq_tbl
as.data.frame(freq_tbl)

###(d) Bar chart (qualitative display)
barplot(freq_tbl,
        main = "Species Composition crabs Dataset",
        xlab = "Species",
        ylab = "Number of crabs",
        col  = c("skyblue", "salmon"))

### e) the data set is balanced accrosss all the groups
```

```
freq_tbl

prop.table(freq_tbl)


## a) boxplot

boxplot(BD ~ sp, data = crabs,

        main = "Body Length by Species",

        xlab = "Species of crabs",

        ylab = "Body length (cm)",

        col  = c("skyblue", "salmon"))


##### b) Group B and O both have outliers but that of O is

### more profound.


###### 3.Carapace length distribution by species

?hist

par(mfrow = c(1, 3))


for (sp in levels(crabs$sp)) {

  x <- crabs$CL[crabs$sp == sp]

  hist(x,

       main = paste("carapace length distribution :",sp),

       xlab = "CL(cm)",

       ylab = "Frequency",

       col  = "gray90",

       border = "yellow")

}
```

```
par(mfrow = c(1, 1))
    border = "yellow"
```

b) Both are unimodal and approximately normal

```
######### comparison of means

    get_mode <- function(x) {
      ux <- unique(x)
      tab <- tabulate(match(x, ux))
      ux[tab == max(tab)]
    }
    ## summary table
    # Overall
    overall <- data.frame(
      Group = "Overall",
      Mean = mean(crabs$FL),
      Median = median(crabs$FL),
      Mode = paste(get_mode(crabs$FL), collapse = ", ")
    )

    # By species

    by_sp <- do.call(rbind, lapply(split(crabs, crabs$Species), function(d) {
```

```
  data.frame(
    Group = as.character(sp$sp[1]),
    Mean = mean(sp$FL),
    Median = median(sp$CL),
    Mode = paste(get_mode(sp$BD), collapse = ", ")
  )
}))

q2_table <- rbind(overall, by_sp)
```